

Boxed-Lunch Ordering Simulation

Business Decision

A catering company must choose between a **Conservative Order** of 230 lunches and a **Large Order** of 270 lunches before actual demand is known. Ordering too few lunches creates lost sales, while ordering too many creates leftover inventory that can only be sold at a discounted salvage value.

Simulation Method

A 1,000-trial Monte Carlo simulation was created in Excel to compare the two ordering policies under uncertain demand and preparation costs.

The model used the following assumptions:

- Selling price per lunch: **\$18**
- Preparation cost: **\$9, \$10, or \$11**, based on the given probabilities
- Salvage value for leftover lunches: **\$5**
- Demand: normally distributed with a mean of **240 lunches** and standard deviation of **45 lunches**
- Demand was rounded to whole lunches and could not fall below zero

For each trial, the model calculated units sold, leftover lunches, lost sales, revenue, preparation cost, salvage revenue, and total profit.

Results

	Conservative	Large
Avg Profit	\$ 1,648.85	\$ 1,656.75
Profit SD	347.7017613	502.7990383
Probability of Loss	0	0.001
Avg Leftover Lunches	12.988	36.697
Avg Lost Sales	22.743	6.452

Recommendation

The recommended policy is the **Large Order**. It produced the highest average profit (\$1,656.75 vs. \$1,648.85) and reduced average lost sales from 22.74 to 6.45 lunches. Although it created more leftovers and had a 0.1% probability of loss, the stronger expected profit and improved ability to meet demand make it the better overall option. Management should revisit this decision if demand, costs, or salvage values change.